Do not write outside the box

0 6 In the Earth most metals are found as compounds.

0 6 . 1 Name **one** metal that is found in the Earth as the metal itself.

[1 mark]

_____

Figure 9 shows a reactivity series.

Figure 9

Most reactive

↑

Potassium

Magnesium

Zinc

Carbon

Metal **Z**

Copper

Least reactive

0 6 . 2 Suggest the most economical method for extracting metal **Z** from an oxide of metal **Z**.

[1 mark]

_____

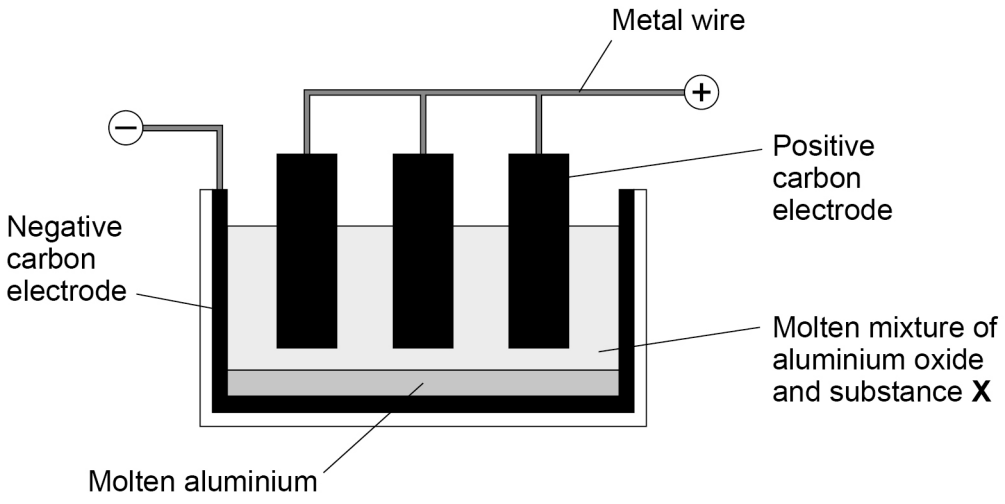
_____

Question 6 continues on the next page

Do not write outside the box

Figure 10 shows the electrolysis cell used to extract aluminium from aluminium oxide.

Figure 10



0 6 . 3 Name substance **X** shown in **Figure 10**.

[1 mark]

_____

Do not write outside the box

0 6 . 4 Explain what happens to the **positive** carbon electrodes during the extraction of aluminium from aluminium oxide.

[3 marks]

_____

_____

_____

_____

_____

_____

0 6 . 5 The formula of aluminium oxide is Al_2O_3

Write a half equation for the reaction at the **negative** electrode in **Figure 10**.

[2 marks]

_____ + _____ → _____

Turn over for the next question

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